

Lab 05

spice

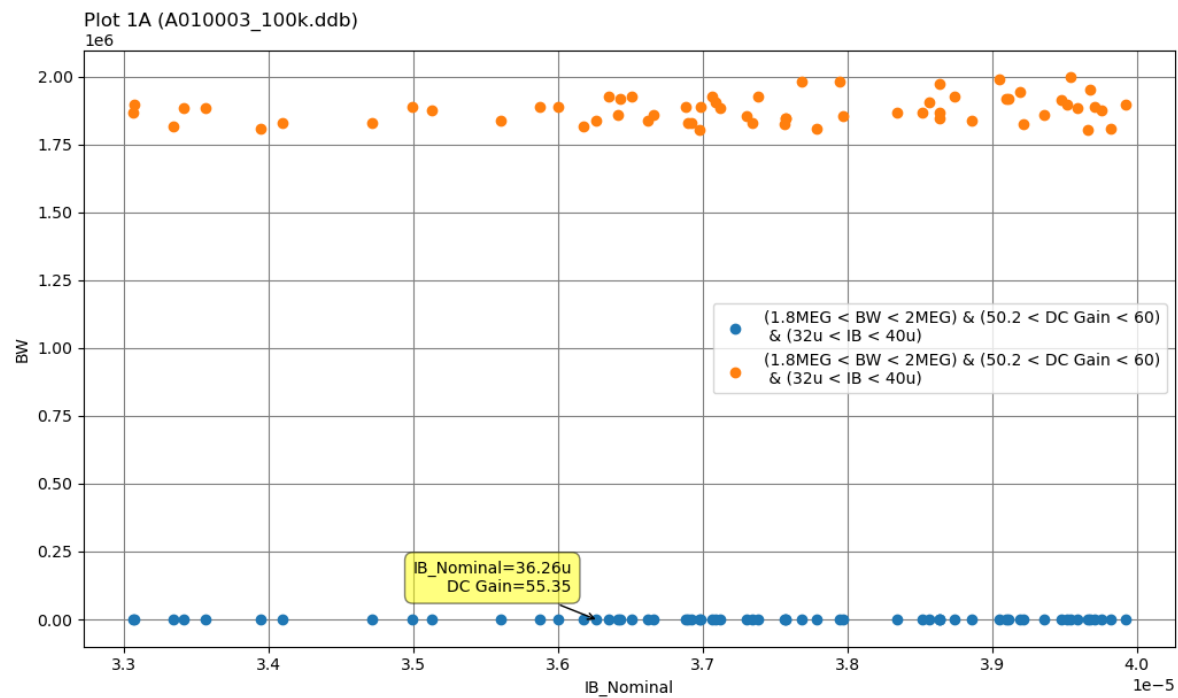
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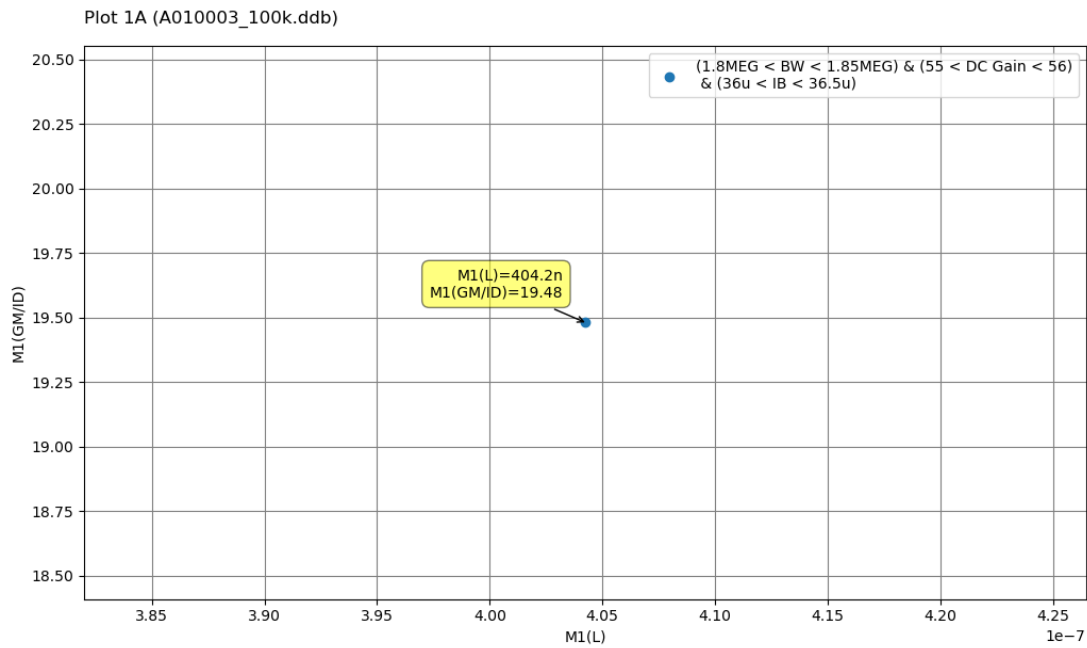
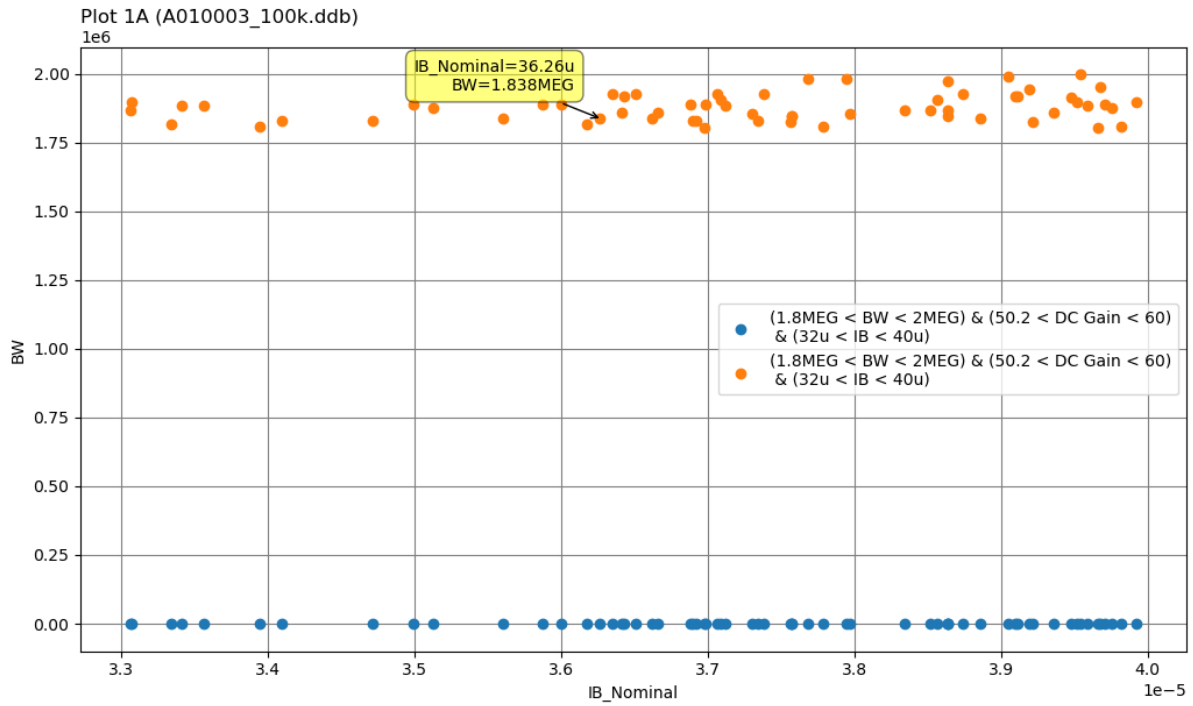
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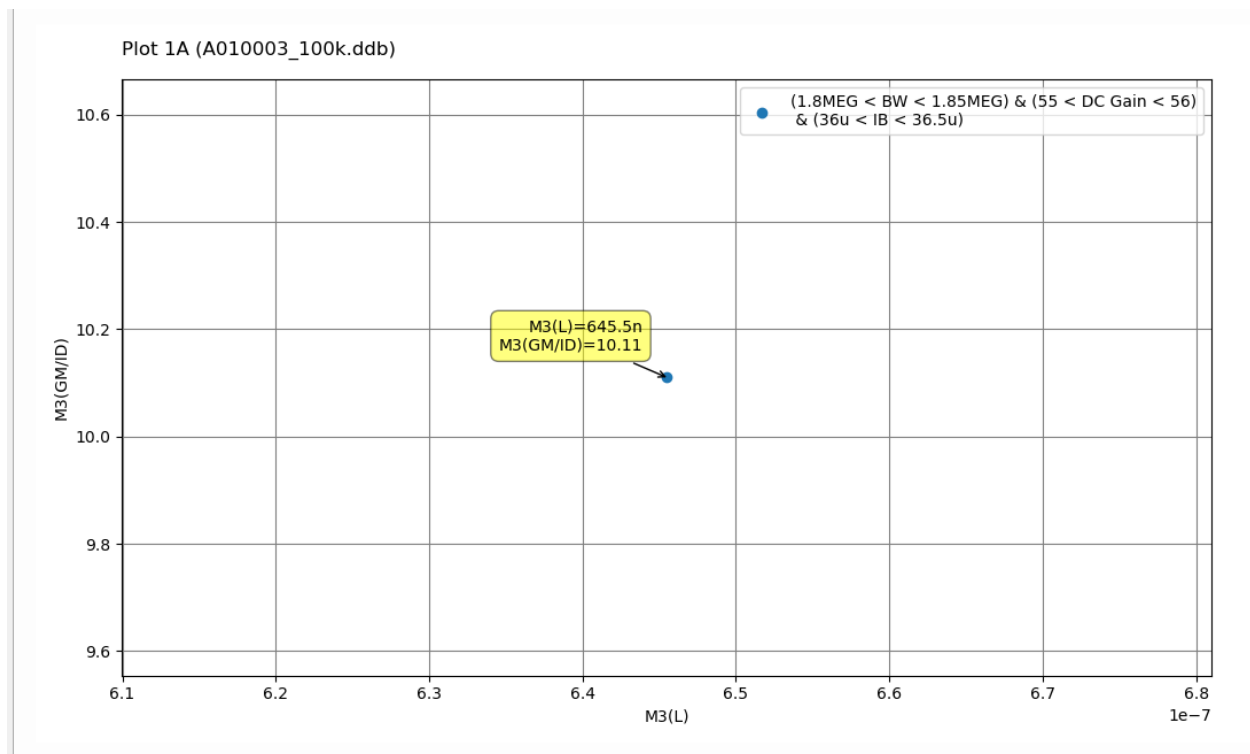
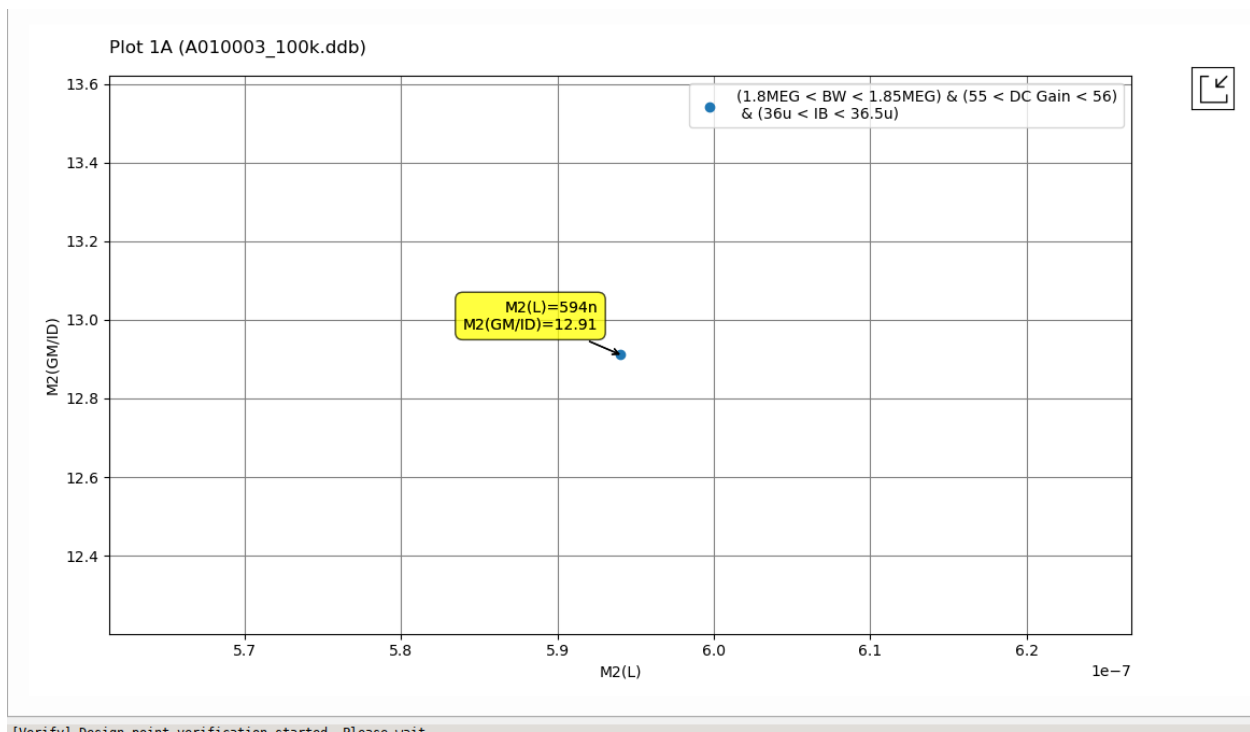
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Part 1 (prelab)

- 1 USE ADT DESIGN XPLORE TO DESIGN AN NMOS-INPUT 5T-OTA, REPORT THE DESIGN CHARTS YOU USED AND USE CURSORS TO CLEARLY SHOW YOUR DESIGN POINTS.

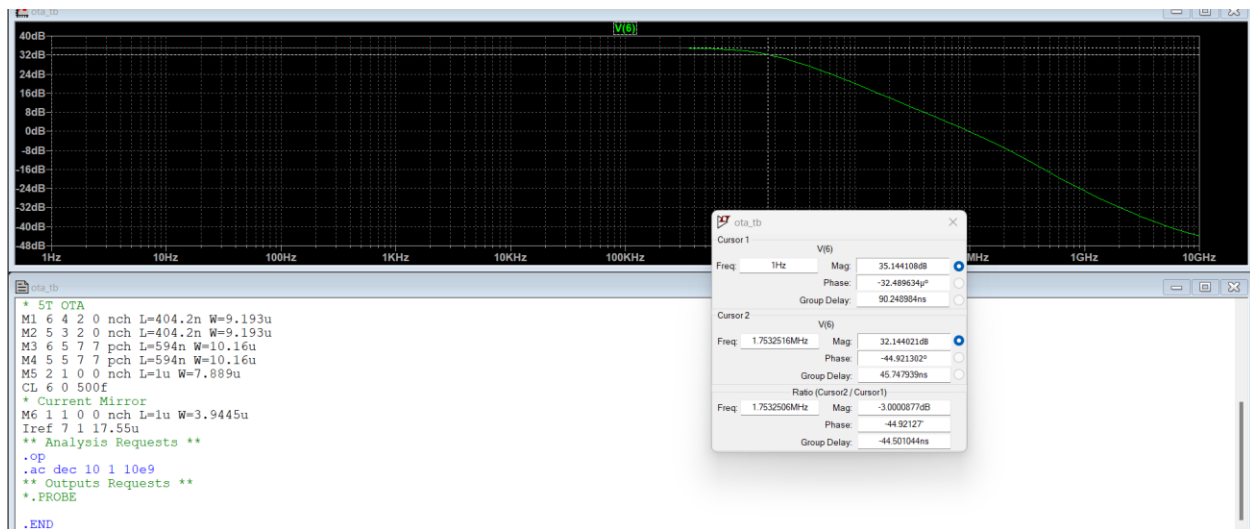




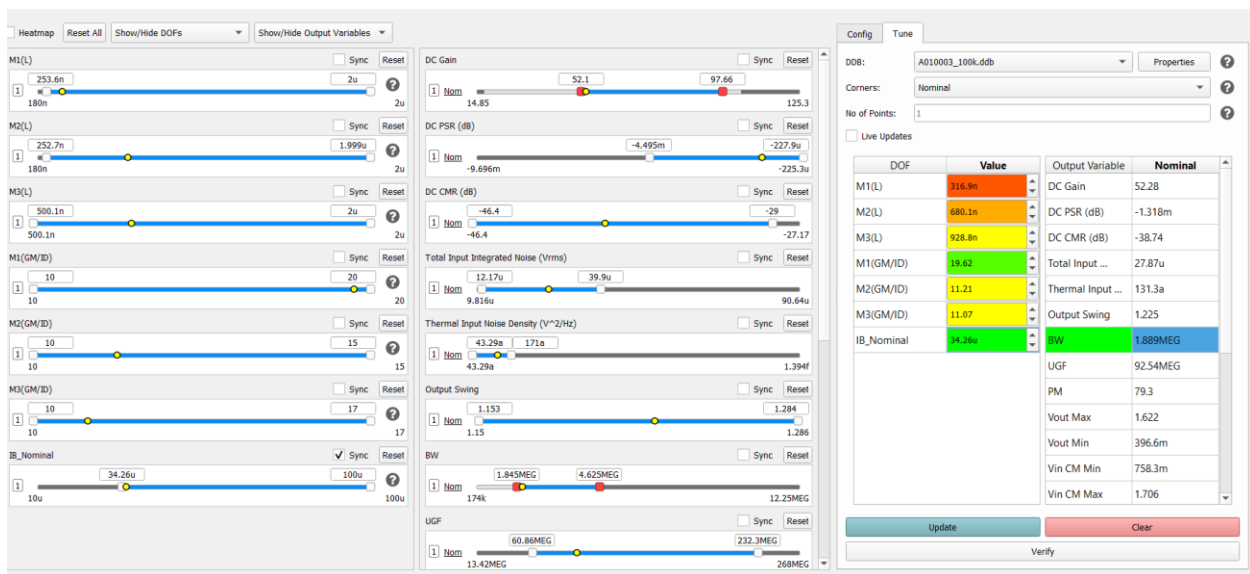


DOF	Value	Output Variable	Nominal
M1(L)	404.2n	DC Gain	57.15
M2(L)	594n	DC PSR (dB)	-1.086m
M3(L)	1u	DC CMR (dB)	-38.8
M1(GM/ID)	19.48	Total Input ...	24.08u
M2(GM/ID)	11	Thermal Inp...	129.2a
M3(GM/ID)	10.11	Output Swing	1.217
IB_Nominal	35.1u	BW	1.76MEG
		UGF	96.16MEG
		PM	80.78
		Vout Max	1.618
		Vout Min	400.7m
		Vin CM Min	774.7m
		Vin CM Max	1.693
		IB	35.1u

Output Variable	Nominal	
	ADT	Sim
DC Gain	57.15	
DC PSR (dB)	-1.086m	
DC CMR (dB)	-38.8	
Total Input Integrated Noise (Vrms)	24.08u	
Thermal Input Noise Density (V^2/Hz)	129.2a	
Output Swing	1.217	
BW	1.76MEG	
UGF	96.16MEG	
PM	80.78	
Vout Max	1.618	
Vout Min	400.7m	
Vin CM Min	774.7m	
Vin CM Max	1.693	
IB	35.1u	
Cgg	30.64f	
$\Sigma W*L$	27.39p	
M1(W)	9.193u	
M2(W)	10.16u	
M3(W)	7.889u	
VBN	612.9m	
Vout	1.194	



2 REPEAT THE PREVIOUS PROBLEM USING ADT DESIGN COCKPIT.
REPORT THE RESULTS.



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DOF	Value	Output Variable	Nominal
M1(L)	317n	DC Gain	51.1
M2(L)	617n	DC PSR (dB)	-1.317m
M3(L)	928.8n	DC CMR (dB)	-38.75
M1(GM/ID)	19.62	Total Input ...	28.17u
M2(GM/ID)	11.21	Thermal Input ...	131.4a
M3(GM/ID)	11.07	Output Swing	1.225
IB_Nominal	34.26u	BW	1.942MEG
		UGF	94.45MEG
		PM	80.64
		Vout Max	1.622
		Vout Min	396.6m
		Vin CM Min	758.3m
		Vin CM Max	1.702

DOF	Value	Output Variable	Nominal
M1(L)	317n	DC Gain	52.16
M2(L)	627n	DC PSR (dB)	-1.482m
M3(L)	928.8n	DC CMR (dB)	-37.74
M1(GM/ID)	19.62	Total Input ...	27.81u
M2(GM/ID)	10	Thermal Input ...	126a
M3(GM/ID)	11.07	Output Swing	1.204
IB_Nominal	34.26u	BW	1.919MEG
		UGF	95.7MEG
		PM	81.13
		Vout Max	1.6
		Vout Min	396.3m
		Vin CM Min	758.6m
		Vin CM Max	1.681

		Nomina
Output Variable	ADT	
DC Gain	52.16	
DC PSR (dB)	-1.482m	
DC CMR (dB)	-37.74	
Total Input Integrated Noise (Vrms)	27.81u	
Thermal Input Noise Density (V ² /Hz)	126a	
Output Swing	1.204	
BW	1.919MEG	
UGF	95.7MEG	
PM	81.13	
Vout Max	1.6	
Vout Min	396.3m	
Vin CM Min	758.6m	
Vin CM Max	1.681	
IB	34.26u	
Cgg	19.87f	
ΣW*L	23.43p	
M1(W)	7.197u	
M2(W)	8.658u	
M3(W)	8.625u	
VBN	598.1m	
Vout	1.177	

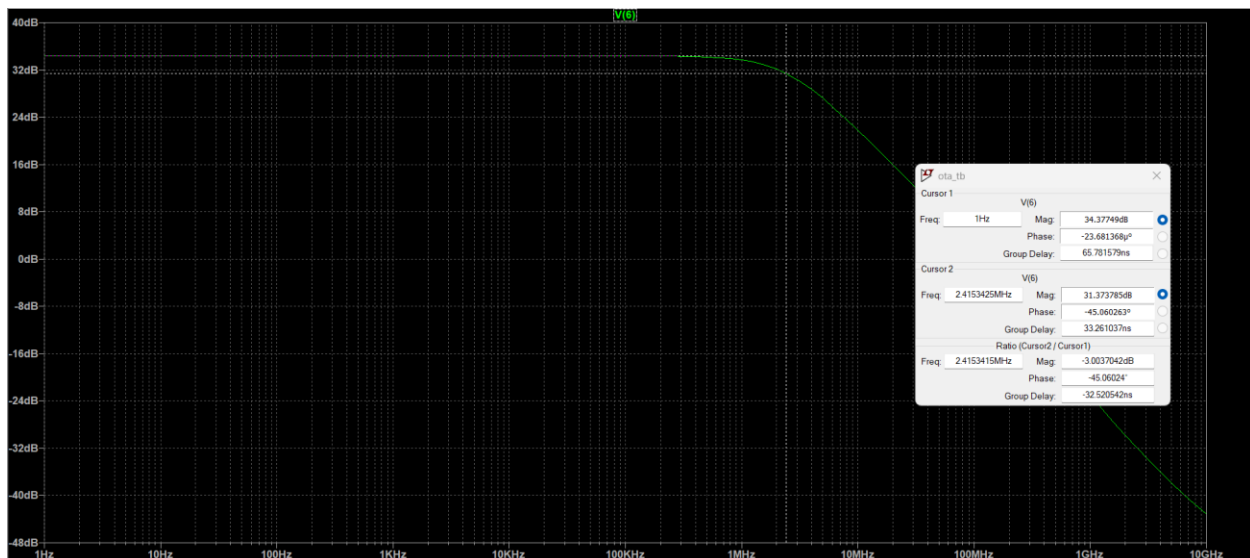
```

5T OTA
* Include external file that contains MOSFET Model
.INCLUDE ee214b_hspice.sp
** Circuit Description **
* power supply
VDD 7 0 DC 1.8
* input
VAC1 3 8 ac 0.5
VAC2 4 9 ac -0.5
VDC1 8 0 dc 900m
VDC2 9 0 dc 900m

* circuit
* 5T OTA
M1 6 4 2 0 nch L=317n W=7.197u
M2 5 3 2 0 nch L=317n W=7.197u
M3 6 5 7 7 pch L=627n W=8.658u
M4 5 5 7 7 pch L=627n W=8.658u
M5 2 1 0 0 nch L=928.8n W=8.625u
CL 6 0 500f
* Current Mirror
M6 1 1 0 0 nch L=1.45u W=4.3125u
Iref 7 1 17.13u
** Analysis Requests **
.op
.ac dec 10 1 10e9
** Outputs Requests **
*.PROBE

.END

```



3 COMPLETE THE FUNCTION “LOPTOTA.M”. REPORT YOUR MATLAB FUNCTION.

```
function OTA = loptOTA(specs)
% inputs in the form of [ M1.L , M3.L , M5.L , M1.RHO ,
M2.RHO , M3.RHO ,
% IB]
OTA = designOTA(specs);
load 180nch.mat;
load 180pch.mat;
UB = [1, 2.00, 1.5 , 20, 15 , 20, 1e-4]; % upper limits
for inputs
LB = [0.18, 0.18, 0.5, 10 , 10 , 10 ,1e-6]; % lower limits
for inputs

%% Add line here for initial inputs (X0 = ??)
X0=[OTA.M1.L,OTA.M3.L,OTA.M5.L,OTA.M1.gm_ID,OTA.M3.gm_ID,OTA.M5.gm_ID,OTA.M5.ID];
%% Add line for the objective function (ObjFn= @ (X) ( ????)
)
ObjFn = @(X) X(end);
NonLinConFn = @(X)NonLinCon(X, OTA, specs,nch,pch); %%
some lines are missing inside NonLinCon file
%% calling fmincon
X = fmincon(ObjFn, X0, [], [], [], [], LB, UB,NonLinConFn);
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% Save all variables
%% Assign X to the corresponding OTA variables (Same lines
as loptOTA)
OTA.M1.L=X(1);
OTA.M3.L=X(2);
OTA.M5.L=X(3);
OTA.M1.gm_ID=X(4);
OTA.M3.gm_ID=X(5);
OTA.M5.gm_ID=X(6);
OTA.M5.ID=X(7);
OTA.M1.ID=X(7)/2 ;
OTA.M3.ID=OTA.M1.ID ;
```

```

OTA.M1.ID_W = look_up(nch, 'ID_W', 'GM_ID', OTA.M1.gm_ID,
'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.W = OTA.M1.ID / OTA.M1.ID_W;
OTA.M3.ID_W = look_up(pch, 'ID_W', 'GM_ID', OTA.M3.gm_ID,
'VDS', OTA.M3.VDS, 'L', OTA.M3.L);
OTA.M3.W = OTA.M3.ID / OTA.M3.ID_W;
OTA.M5.ID_W = look_up(nch, 'ID_W', 'GM_ID', OTA.M5.gm_ID,
'VDS', OTA.M5.VDS, 'L', OTA.M5.L);
OTA.M5.W = OTA.M5.ID / OTA.M5.ID_W;
OTA.M1.VGS = look_upVGS(nch, 'GM_ID', OTA.M1.gm_ID,
'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.VG = OTA.M1.VGS + OTA.M5.VDS;

```

3.1 NonLinCon

```

function [C, CEQ] = NonLinCon(X, OTA, specs,nch,pch)

%% add lines here relating OTA variables to X
%%
OTA.M1.L=X(1);
OTA.M3.L=X(2);
OTA.M5.L=X(3);
OTA.M1.gm_ID=X(4);
OTA.M3.gm_ID=X(5);
OTA.M5.gm_ID=X(6);
OTA.M5.ID=X(7);
OTA.M1.ID=X(7)/2;
OTA.M3.ID=OTA.M1.ID;

OTA.M1.gm = OTA.M1.ID*OTA.M1.gm_ID;
OTA.M3.gm = OTA.M1.ID*OTA.M3.gm_ID;

OTA.M1.gm_gds=look_up(nch, 'GM_GDS', 'GM_ID', OTA.M1.gm_ID,
'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.gds = OTA.M1.gm / OTA.M1.gm_gds;
OTA.M3.gm_gds=look_up(pch, 'GM_GDS', 'GM_ID', OTA.M3.gm_ID,
'VDS', OTA.M3.VDS, 'L', OTA.M3.L);
OTA.M3.gds = OTA.M3.gm / OTA.M3.gm_gds;
OTA.M1.gm_cdd = look_up(nch, 'GM_CDD', 'GM_ID',
OTA.M1.gm_ID, 'VDS', OTA.M1.VDS, 'L', OTA.M1.L);

```



```

%% Assign X to the corresponding OTA variables , for
example:
OTA.M1.L=X(1);
OTA.M3.L=X(2);
OTA.M5.L=X(3);
OTA.M1.gm_ID=X(4);
OTA.M3.gm_ID=X(5);
OTA.M5.gm_ID=X(6);
OTA.M5.ID=X(7);
OTA.M1.ID=OTA.M5.ID/2 ;
OTA.M3.ID=OTA.M1.ID ;

%% continue the rest (multiple lines)
%% note that if you did not write the lines , the values
from the previous lab will be used automaticlly
OTA.M1.ID_W = look_up(nch, 'ID_W', 'GM_ID', OTA.M1.gm_ID,
'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.W = OTA.M1.ID / OTA.M1.ID_W;
OTA.M3.ID_W = look_up(pch, 'ID_W', 'GM_ID', OTA.M3.gm_ID,
'VDS', OTA.M3.VDS, 'L', OTA.M3.L);
OTA.M3.W = OTA.M3.ID / OTA.M3.ID_W;
OTA.M5.ID_W = look_up(nch, 'ID_W', 'GM_ID', OTA.M5.gm_ID,
'VDS', OTA.M5.VDS, 'L', OTA.M5.L);
OTA.M5.W = OTA.M5.ID / OTA.M5.ID_W;
OTA.M1.VGS = look_upVGS(nch, 'GM_ID', OTA.M1.gm_ID, 'VDS',
OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.VG = OTA.M1.VGS + OTA.M5.VDS;

```

4.1 NonLinCONV

```
function [C, CEQ] = NonLinConV(X, OTA, specs,nch,pch)
```

```

OTA.M1.L = X(:,1);
OTA.M3.L = X(:,2);
OTA.M5.L = X(:,3);
OTA.M1.gm_ID = X(:,4);
OTA.M3.gm_ID = X(:,5);
OTA.M5.gm_ID = X(:,6);
OTA.M5.ID = X(:,7);
OTA.M1.ID = 0.5*X(:,7);

```

```

OTA.M3.ID = 0.5*X(:,7);

OTA.M1.gm = OTA.M1.ID.*OTA.M1.gm_ID;
OTA.M3.gm = OTA.M1.ID.*OTA.M3.gm_ID;

OTA.M1.gm_gds=look_up(nch, 'GM_GDS', 'GM_ID', OTA.M1.gm_ID,
'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.gm_gds=diag(OTA.M1.gm_gds);
OTA.M1.gds = OTA.M1.gm ./ OTA.M1.gm_gds;
OTA.M3.gm_gds=look_up(pch, 'GM_GDS', 'GM_ID', OTA.M3.gm_ID,
'VDS', OTA.M3.VDS, 'L', OTA.M3.L);
OTA.M3.gm_gds=diag(OTA.M3.gm_gds);
OTA.M3.gds = OTA.M3.gm ./ OTA.M3.gm_gds;
OTA.M1.gm_cdd = look_up(nch, 'GM_CDD', 'GM_ID',
OTA.M1.gm_ID, 'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.gm_cdd =diag(OTA.M1.gm_cdd);
OTA.M3.gm_cdd = look_up(pch, 'GM_CDD', 'GM_ID',
OTA.M3.gm_ID, 'VDS', OTA.M3.VDS, 'L', OTA.M3.L);
OTA.M3.gm_cdd =diag(OTA.M3.gm_cdd);
OTA.M1.cdd=OTA.M1.gm ./ OTA.M1.gm_cdd;
OTA.M3.cdd=OTA.M3.gm ./ OTA.M3.gm_cdd;

%% write ouput specs AVDC_OUT , GBW_OUT equations (include
self loading )
C_total = OTA.M1.cdd+OTA.M3.cdd+specs.CL;
GBW_OUT=OTA.M1.gm ./ (2*pi.*C_total);
AVDC_OUT=OTA.M1.gm./ (OTA.M1.gds+OTA.M3.gds) ;

CEQ = [];
C(:,1) = 10^(specs.AVDC / 20) ./ double(AVDC_OUT) - 1;
C(:,2) = specs.GBW ./ double(GBW_OUT) - 1;

```

5 COMPLETE THE SCRIPT “OPTOTA_TEST.M”, REPORT YOUR SCRIPT. REPORT THE OUTPUT OF THE COMMAND WINDOW. COMPARE ALL THE RESULTS FROM ADT AND MATLAB IN A TABLE. COMMENT ON THE RESULTS

```
% OTA Design Script
% Write the SPECS
clear all;
AVDC = 34; %dB
GBW = 1e8; %Hz
CL = 500e-15; %Farad
specs = struct('AVDC', AVDC,...
'CL', CL,...
'GBW', GBW);

OTA1 = loptOTA(specs);
% Print the solution
fprintf('**** Local opt OTA Design ****\n\n');
fprintf('Input Pair:\n');
fprintf('      L = %.2f um\n      W=%.2f um\n      ViCM=%.4f\n',OTA1.M1.L,OTA1.M1.W,OTA1.M1.VG);
fprintf('CM Load:\n');
fprintf('      L = %.2f um\n      W=%.2f um\n',OTA1.M3.L,OTA1.M3.W);
fprintf('Tail Current Source:\n');
fprintf('      L = %.2f um\n      W=%.2f um\n\n',OTA1.M5.L,OTA1.M5.W);

OTA = goptOTA(specs);
fprintf('**** global opt OTA Design ****\n\n');
fprintf('Input Pair:\n');
fprintf('      L = %.2f um\n      W=%.2f um\n      ViCM=%.4f\n\n',OTA.M1.L,OTA.M1.W,OTA.M1.VG);
fprintf('CM Load:\n');
fprintf('      L = %.2f um\n      W=%.2f um\n\n',OTA.M3.L,OTA.M3.W);
ID=OTA.M3.ID
fprintf('Tail Current Source:\n');
fprintf('      L = %.2f um\n      W=%.2f um\n\n',OTA.M5.L,OTA.M5.W);
%%
```

**** Local opt OTA Design ****

Input Pair:

L = 0.42 um
W=11.10 um
ViCM=1.1159 V

CM Load:

L = 0.52 um
W=8.35 um

Tail Current Source:

L = 1.00 um
W=19.17 um

**** global opt OTA Design ****

Input Pair:

L = 0.40 um
W=11.71 um
ViCM=1.1159 V

CM Load:

L = 1.01 um
W=15.57 um

Tail Current Source:

L = 1.45 um
W=73.95 um

OTAM5

Field	Value
L	1.0000
ID	3.3578e-05
VDS	0.6000
gm_ID	15.0001
ID_W	1.7516e-06
W	19.1696

Field	Value
L	1.4488
ID	3.5742e-05
VDS	0.6000
gm_ID	19.1176
ID_W	4.8334e-07
W	73.9491

		ADT Design Xplore	ADT Design Cockpit	local optimization	global optimization
M1(in)	L (um)	0.404	0.317	0.42	0.40
	W (um)	9.193	7.197	11.10	11.71
M3(Load)	L (um)	0.594	0.627	0.52	1.01
	W (um)	10.16	8.658	8.35	15.57
M5(TAIL)	L (um)	1	0.928	1	1.45
	W (um)	7.889	4.336	19.17	73.95
	IB	35.1	34.26	33.57	35.74

Global optimization changes every time we run the function, the result is not accurate, as the global optimization should give more optimized result than local optimization case, but that requires increasing 'PopulationSize' and decreasing 'FunctionTolerance' which leads to exhausting simulation time.

The differences between Design Xplore and Design cockpit are due to differences in design procedures and approaches, surly also the designer proficiency of using the tool.

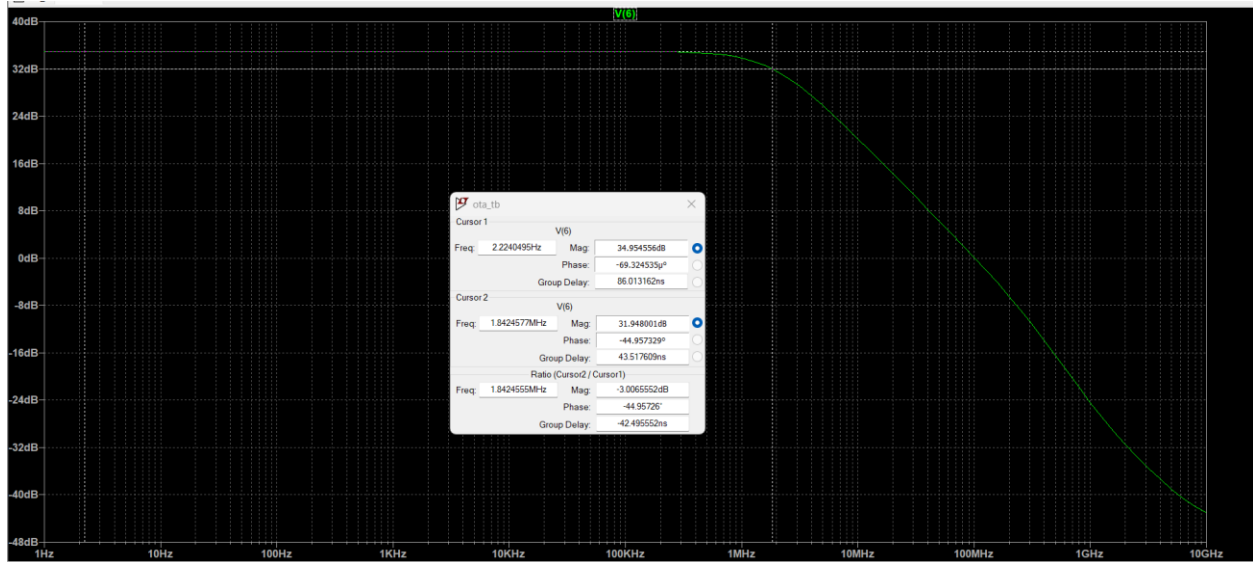
6 COMPLETE THE NETLIST “OTA_TB.CIR” TO SIMULATE YOUR SYNTHESIZED OTA. COMPARE THE SIMULATOR OUTPUT WITH THE REQUIRED SPECIFICATIONS. REPORT YOUR NETLIST, THE SIMULATION RESULTS, AND THE COMPARISON TABLE. COMMENT ON THE RESULTS.

Local:

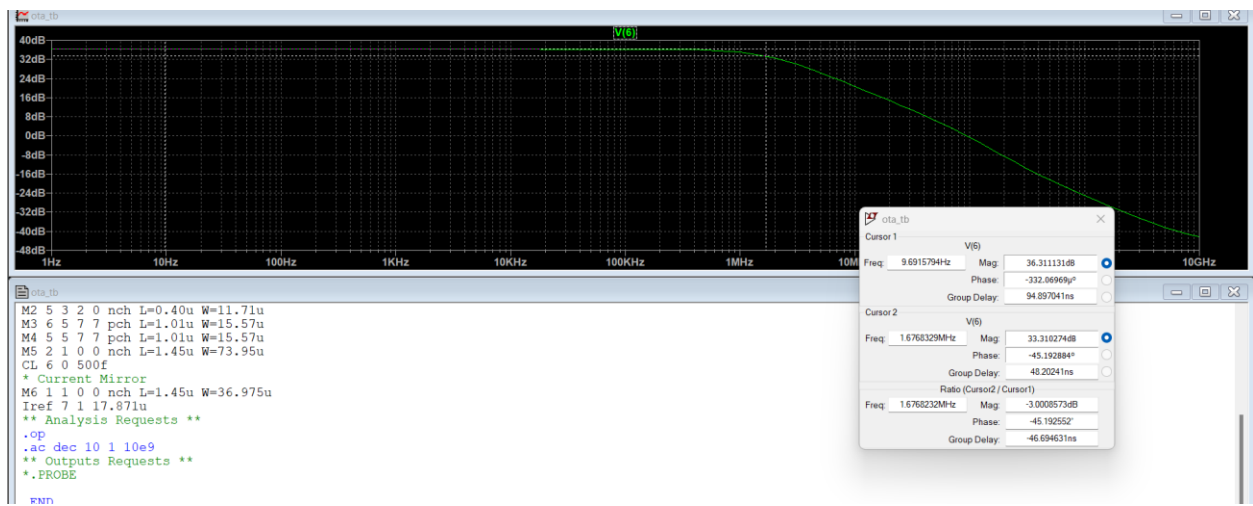
```
5T OTA
* Include external file that contains MOSFET Model
.INCLUDE ee214b_hspice.sp
** Circuit Description **
* power supply
VDD 7 0 DC 1.8
* input
VAC1 3 8 ac 0.5
VAC2 4 9 ac -0.5
VDC1 8 0 dc 1.1159
VDC2 9 0 dc 1.1159

* circuit
* 5T OTA
M1 6 4 2 0 nch L=0.42u W=11.10u
M2 5 3 2 0 nch L=0.42u W=11.10u
M3 6 5 7 7 pch L=0.52u W=8.35u
M4 5 5 7 7 pch L=0.52u W=8.35u
M5 2 1 0 0 nch L=1u W=19.17u
CL 6 0 500f
* Current Mirror
M6 1 1 0 0 nch L=1u W=9.585u
Iref 7 1 16.789u
** Analysis Requests **
.op
.ac dec 10 1 10e9
** Outputs Requests **
*.PROBE

.END
```



Global:



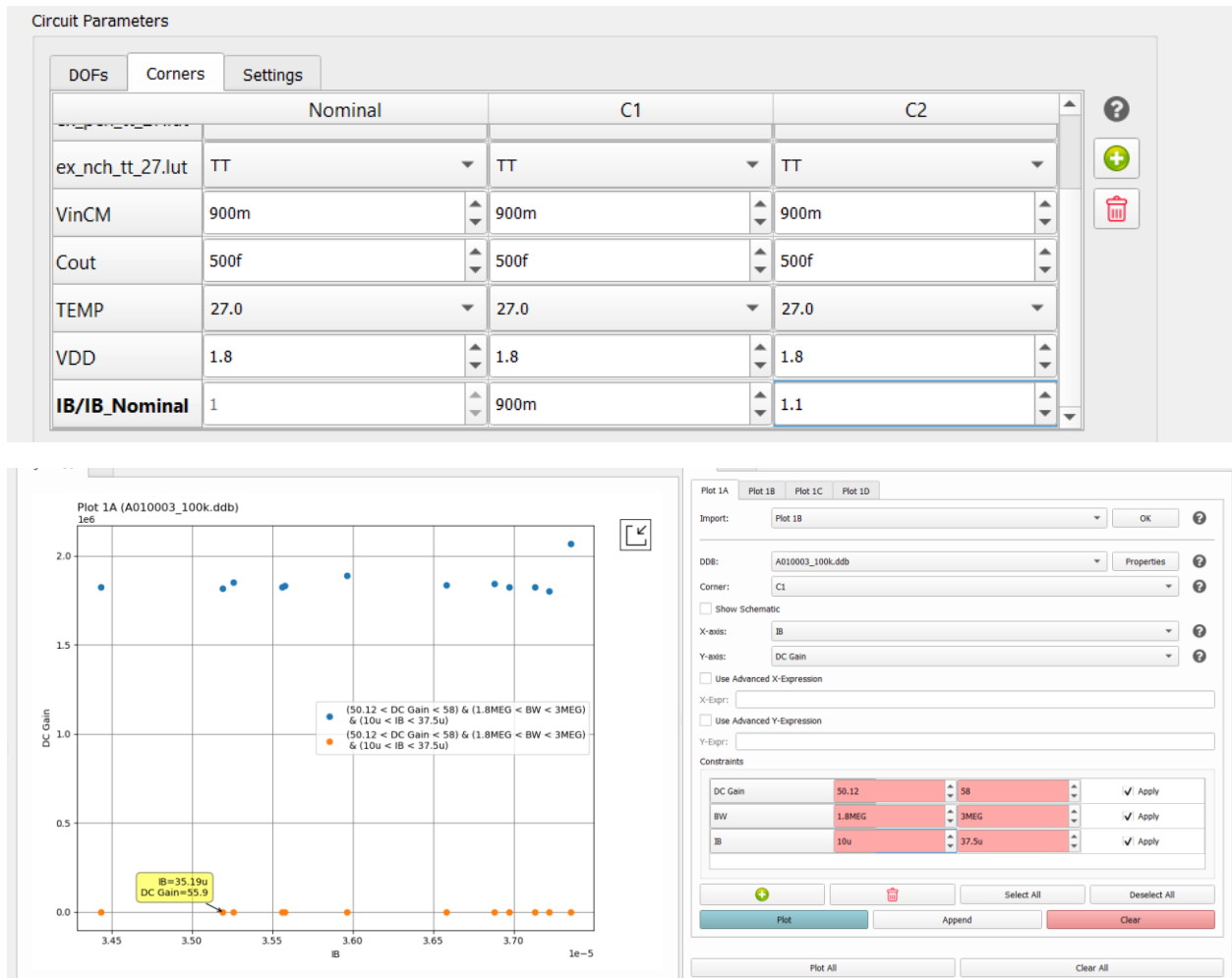
6.1 COMPARISON TABLE

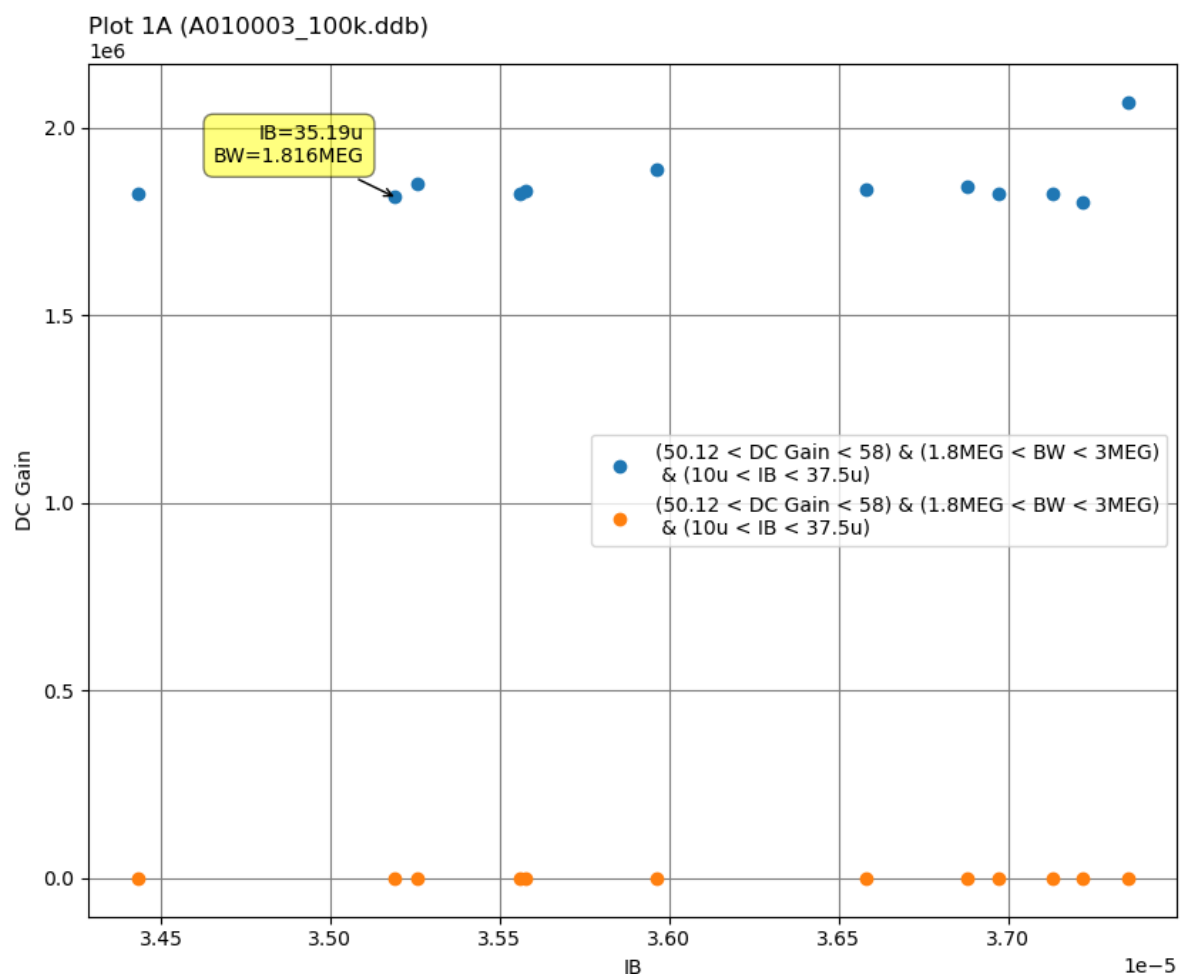
	SPICE(adt)	Spice(cockpit)	Spice(local)	Global	SPECS
GBW	50.44*2.01 =101.4 MHZ	52.36*2.41 =126.18 MHZ	55.91*1.84 =102.87MHZ	65.31*1.678 =109.59 MHZ	100 MHZ
GAIN	34.056 DB	34.38 DB	34.95 DB	36.3 DB	34 DB
IB	35.1 A	34.26 A	33.58 A	35.74 A	

Results meet the specs, we have higher GBW and gain than our specs and in the same time we use less bias current than we had in the previous lab, global optimization code is not practical and need a severe simulation time to work accurately.

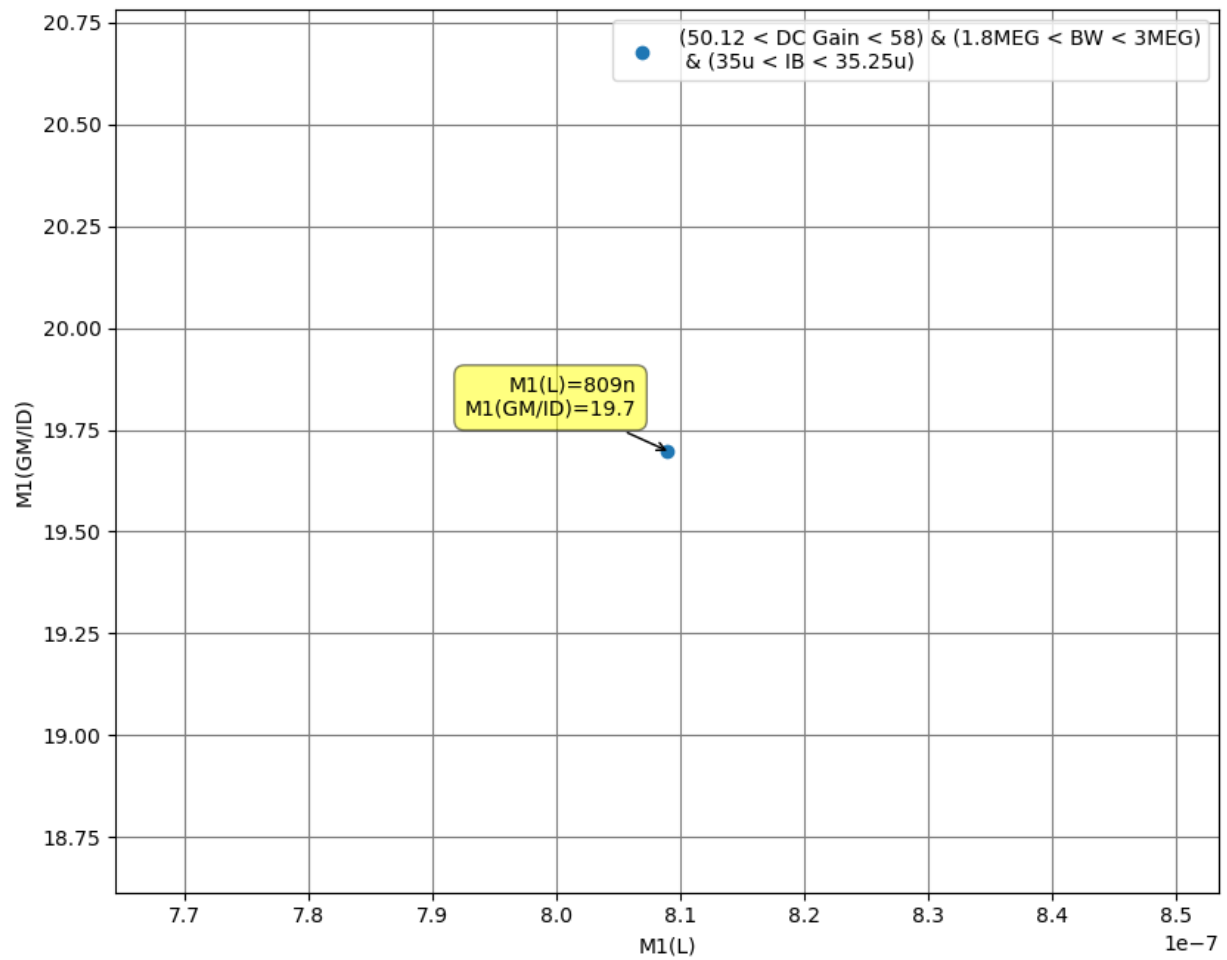
PART2

1 ADT DESIGN XPLORE.

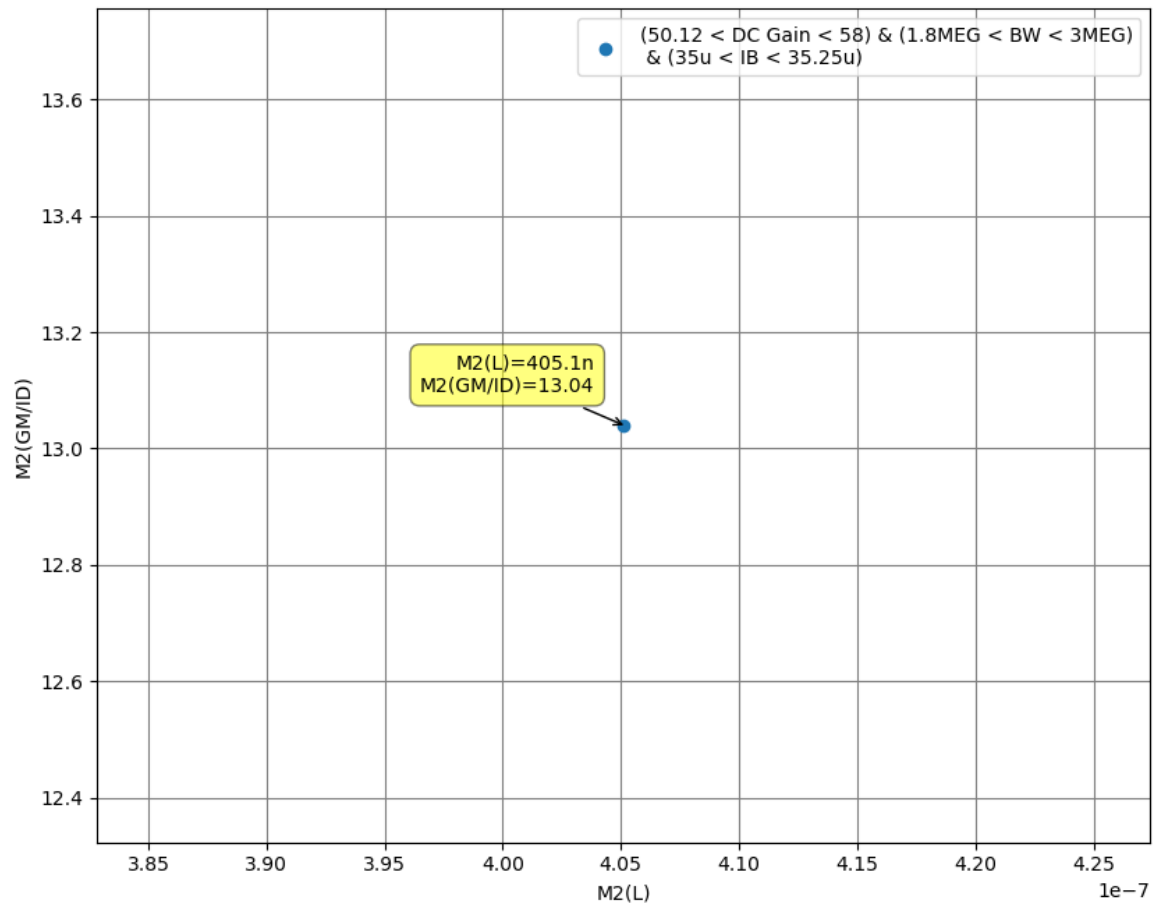


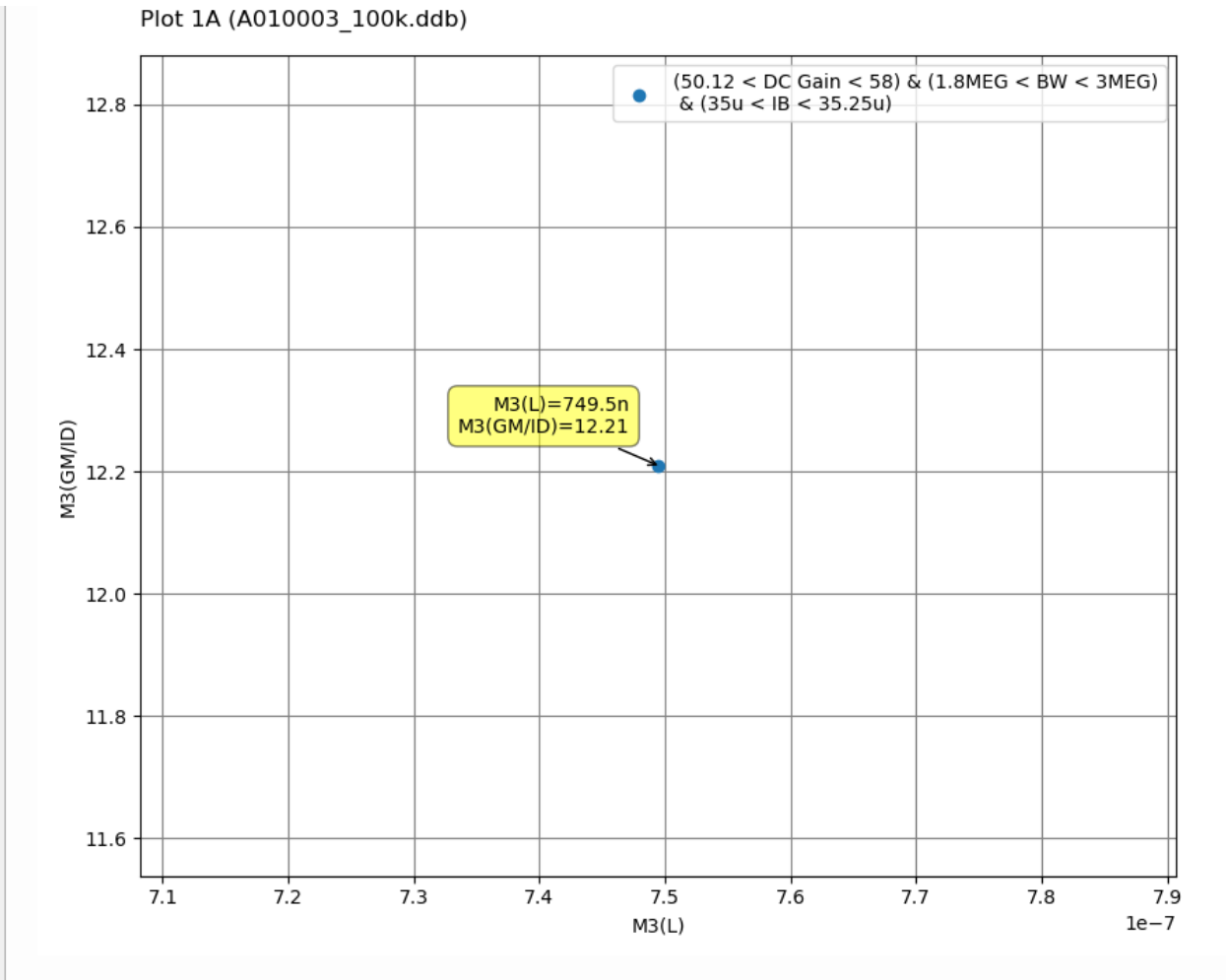


Plot 1A (A010003_100k.ddb)



Plot 1A (A010003_100k.ddb)





DOF	Value	Output Variable	Nominal	C1	C2
M1(L)	809n	DC Gain	55.89	55.92	55.84
M2(L)	405.1n	DC PSR (dB)	-689u	-656u	-719.2u
M3(L)	749.5n	DC CMR (dB)	-40.44	-40.88	-40.06
M1(GM/ID)	19.7	Total Input ...	21.48u	21.5u	21.44u
M2(GM/ID)	13.04	Thermal Input ...	123.5a	134.2a	114.1a
M3(GM/ID)	12.21	Output Swing	1.22	1.223	1.215
IB_Nominal	38.7u	BW	1.942MEG	1.795MEG	2.093MEG
		UGF	105.4MEG	97.7MEG	113.4MEG
		PM	81.97	82.08	81.87
		Vout Max	1.647	1.652	1.639
		Vout Min	426.5m	429.5m	424.9m
		Vin CM Min	697.1m	683.1m	711.9m
		Vin CM Max	1.687	1.692	1.681

☐ Live Updates

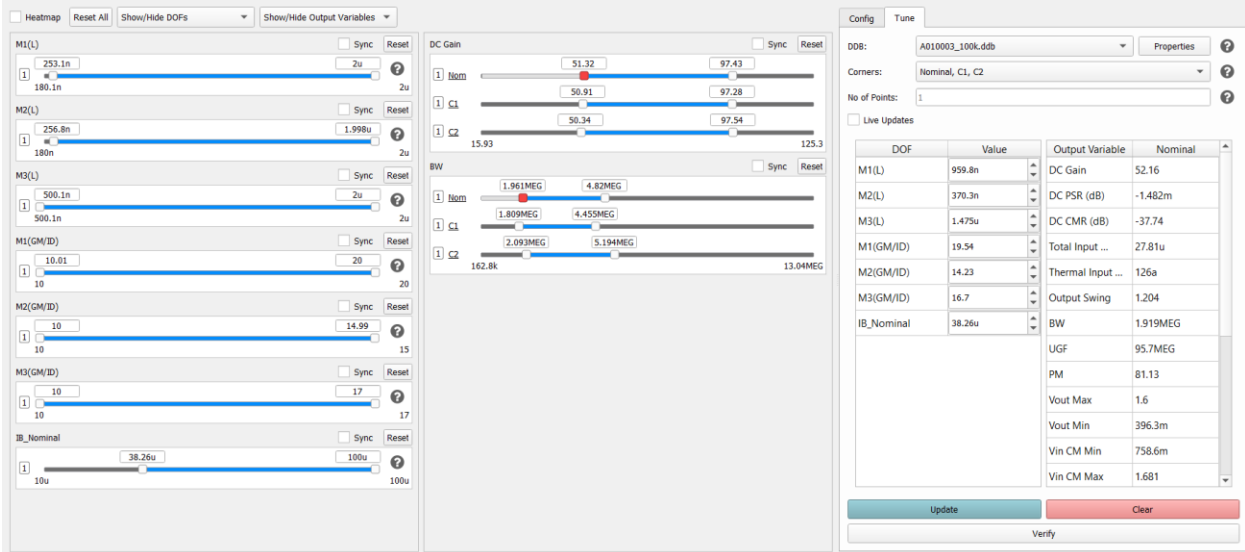
DOF	Value	Output Variable	Nominal	C1	C2
M1(L)	809n	DC Gain	55.89	55.92	55.84
M2(L)	405.1n	DC PSR (dB)	-689u	-656u	-719.2u
M3(L)	749.5n	DC CMR (dB)	-40.44	-40.88	-40.06
M1(GM/ID)	19.7	Total Input ...	21.54u	21.56u	21.5u
M2(GM/ID)	13.04	Thermal Input ...	125.8a	136.6a	116.2a
M3(GM/ID)	12.21	Output Swing	1.22	1.223	1.215
IB_Nominal	38u	BW	1.912MEG	1.767MEG	2.06MEG
		UGF	103.8MEG	96.18MEG	111.7MEG
		PM	82.09	82.21	81.99
		Vout Max	1.647	1.652	1.639
		Vout Min	426.5m	429.5m	424.9m
		Vin CM Min	697.1m	683.1m	711.9m
		Vin CM Max	1.687	1.692	1.681

DOF	Value	Output Variable	Nominal	C1	C2
M1(L)	789n	DC Gain	55.65	55.69	55.61
M2(L)	405.1n	DC PSR (dB)	-697.2u	-663u	-728u
M3(L)	749.5n	DC CMR (dB)	-40.41	-40.86	-40.03
M1(GM/ID)	19.7	Total Input ...	21.63u	21.68u	21.58u
M2(GM/ID)	13.04	Thermal Input ...	125.9a	136.7a	116.2a
M3(GM/ID)	12.21	Output Swing	1.221	1.223	1.216
IB_Nominal	38u	BW	1.919MEG	1.781MEG	2.068MEG
		UGF	103.8MEG	96.33MEG	111.7MEG
		PM	82.16	82.27	82.07
		Vout Max	1.647	1.652	1.639
		Vout Min	425.6m	428.6m	423.9m
		Vin CM Min	698.3m	684.2m	713m
		Vin CM Max	1.688	1.693	1.682

DOF	Value	Output Variable	Nominal	C1	C2
M1(L)	789n	DC Gain	56.98	56.93	56.94
M2(L)	400n	DC PSR (dB)	-755.4u	-717.7u	-791.9u
M3(L)	749.5n	DC CMR (dB)	-39.73	-40.18	-39.31
M1(GM/ID)	19.8	Total Input ...	21.31u	21.38u	21.26u
M2(GM/ID)	12	Thermal Input ...	120.9a	131.4a	111.4a
M3(GM/ID)	12.21	Output Swing	1.207	1.21	1.201
IB_Nominal	38u	BW	1.897MEG	1.76MEG	2.044MEG
		UGF	105.4MEG	97.75MEG	113.3MEG
		PM	82.48	82.6	82.37
		Vout Max	1.633	1.639	1.625
		Vout Min	426m	429m	424.4m
		Vin CM Min	697.1m	683.1m	711.9m
		Vin CM Max	1.672	1.677	1.665

Output Variable	Nominal	
	ADT	Simulator
DC Gain	56.98	
DC PSR (dB)	-755.4u	
DC CMR (dB)	-39.73	
Total Input Integrated Noise (Vrms)	21.31u	
Thermal Input Noise Density (V ² /Hz)	120.9a	
Output Swing	1.207	
BW	1.897MEG	
UGF	105.4MEG	
PM	82.48	
Vout Max	1.633	
Vout Min	426m	
Vin CM Min	697.1m	
Vin CM Max	1.672	
IB	38u	
Cgg	124.2f	
ΣW*L	48.23p	
M1(W)	21.84u	
M2(W)	8.356u	
M3(W)	9.455u	
VBN	589.4m	
Vout	1.198	

2 ADT DESIGN COCKPIT



we used the previous point in part1 and tuned, our aim to get less current than 38.26 uA which is what expected from our reasonable assumed constraints.

DOF	Value	Output Variable	Nominal	C1	C2
M1(L)	317n	DC Gain	52.16	51.93	52.16
M2(L)	627n	DC PSR (dB)	-1.481m	-1.364m	-1.621m
M3(L)	930n	DC CMR (dB)	-37.75	-38.47	-36.95
M1(GM/ID)	19.62	Total Input ...	27.65u	27.68u	27.6u
M2(GM/ID)	10	Thermal Input ...	123.4a	136.1a	113.3a
M3(GM/ID)	11.07	Output Swing	1.204	1.21	1.193
IB_Nominal	35u	BW	1.957MEG	1.802MEG	2.109MEG
		UGF	97.49MEG	89.67MEG	105MEG
		PM	80.98	81.24	80.79
		Vout Max	1.6	1.609	1.588
		Vout Min	396.3m	398.7m	395.3m
		Vin CM Min	758.6m	743.5m	775.3m
		Vin CM Max	1.681	1.688	1.671

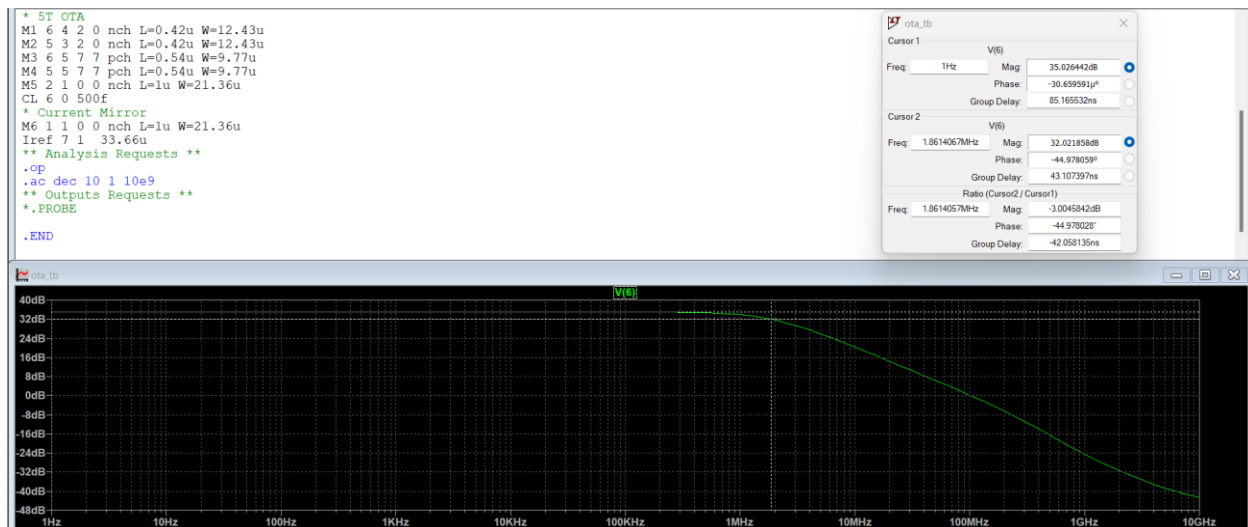
DOF	Value	Output Variable	Nominal	C1	C2
M1(L)	317n	DC Gain	52.22	51.96	52.25
M2(L)	627n	DC PSR (dB)	-1.472m	-1.357m	-1.611m
M3(L)	930n	DC CMR (dB)	-37.8	-38.52	-37.01
M1(GM/ID)	19.72	Total Input ...	26.99u	27.02u	26.96u
M2(GM/ID)	10	Thermal Input ...	114.4a	126.5a	104.9a
M3(GM/ID)	11.07	Output Swing	1.203	1.21	1.193
IB_Nominal	37.5u	BW	2.093MEG	1.927MEG	2.264MEG
		UGF	103.9MEG	95.5MEG	112.1MEG
		PM	80.39	80.67	80.18
		Vout Max	1.6	1.609	1.588
		Vout Min	396.8m	399m	395.8m
		Vin CM Min	757.3m	742.3m	774m
		Vin CM Max	1.68	1.688	1.671

Output Variable	ADT
DC Gain	52.22
DC PSR (dB)	-1.472m
DC CMR (dB)	-37.8
Total Input Integrated Noise (Vrms)	26.99u
Thermal Input Noise Density (V ² /Hz)	114.4a
Output Swing	1.203
BW	2.093MEG
UGF	103.9MEG
PM	80.39
Vout Max	1.6
Vout Min	396.8m
Vin CM Min	757.3m
Vin CM Max	1.68
IB	37.5u
Cgg	22.25f
$\Sigma W * L$	25.8p
M1(W)	8.081u
M2(W)	9.476u
M3(W)	9.451u
VBN	598m
Vout	1.177

3 MATLAB LOCAL OPTIMIZATION (FMINCON).

For worst case corner:

```
Editor - D:\spice\DONE\LAB5\NonLinCon.m
optOTA_test.m x loptOTA.m x NonLinCon.m x +
1
2 function [C, CEQ] = NonLinCon(X, OTA, specs,nch,pch)
3 %cc=[1.1,1,0.9];
4
5 %% add lines here relating OTA variables to X
6 %%
7 - OTA.M1.L=X(1);
8 - OTA.M3.L=X(2);
9 - OTA.M5.L=X(3);
10 - OTA.M1.gm_ID=X(4);
11 - OTA.M3.gm_ID=X(5);
12 - OTA.M5.gm_ID=X(6);
13 - OTA.M5.ID=0.9*X(7);
14 - OTA.M1.ID=OTA.M5.ID/2 ;
15 - OTA.M3.ID=OTA.M1.ID ;
16
**** Local opt OTA Design ****
Input Pair:
    L = 0.42 um
    W=12.43 um
    ViCM=1.1157 V
CM Load:
    L = 0.54 um
    W=9.77 um
Tail Current Source:
    L = 1.00 um
    W=21.36 um
|
IB =
3.7432e-05
```



Achieved the specs at the worst-case corner.

We can change the constraints and try all the corners:

```
function [C, CEQ] = NonLinCon(X, OTA, specs,nch,pch)
cc=[1.1,1,0.9];
%% add lines here relating OTA variables to X
for i =1:3
OTA.M1.L=X(1);
OTA.M3.L=X(2);
OTA.M5.L=X(3);
OTA.M1.gm_ID=X(4);
OTA.M3.gm_ID=X(5);
OTA.M5.gm_ID=X(6);
OTA.M5.ID=X(7).*cc(i);
OTA.M1.ID=OTA.M5.ID/2;
OTA.M3.ID=OTA.M1.ID;
OTA.M1.gm = OTA.M1.ID*OTA.M1.gm_ID;
OTA.M3.gm = OTA.M1.ID*OTA.M3.gm_ID
OTA.M1.gm_gds=look_up(nch, 'GM_GDS', 'GM_ID', OTA.M1.gm_ID,
'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M1.gds = OTA.M1.gm / OTA.M1.gm_gds;
OTA.M3.gm_gds=look_up(pch, 'GM_GDS', 'GM_ID', OTA.M3.gm_ID,
'VDS', OTA.M3.VDS, 'L', OTA.M3.L);
OTA.M3.gds = OTA.M3.gm / OTA.M3.gm_gds;
OTA.M1.gm_cdd = look_up(nch, 'GM_CDD', 'GM_ID',
OTA.M1.gm_ID, 'VDS', OTA.M1.VDS, 'L', OTA.M1.L);
OTA.M3.gm_cdd = look_up(pch, 'GM_CDD', 'GM_ID',
OTA.M3.gm_ID, 'VDS', OTA.M3.VDS, 'L', OTA.M3.L);
```

```

OTA.M1.cdd=OTA.M1.gm/OTA.M1.gm_cdd;
OTA.M3.cdd=OTA.M3.gm/OTA.M3.gm_cdd;
C_total = OTA.M1.cdd+OTA.M3.cdd+specs.CL ;
%% add lines here to calculate specs AVDC_OUT , GBW_OUT
GBW_OUT=( OTA.M1.gm ./ (2.*pi.*C_total) );
AVDC_OUT=(OTA.M1.gm / (OTA.M1.gds+OTA.M3.gds)) ;
CEQ = [];
C(2*i) = 10^(specs.AVDC / 20) / double(AVDC_OUT) - 1;
C(2*i-1) = specs.GBW / double(GBW_OUT) - 1;
end

```

Input Pair:

L = 0.48 um

W=14.45 um

ViCM=1.1093 V

CM Load:

L = 0.64 um

W=11.67 um

Tail Current Source:

L = 1.00 um

W=21.58 um

IB =

3.7801e-05

It also gives very close results to the worst case only, but in the first code we had more optimized current.

4 MATLAB GENETIC OPTIMIZATION

For worst case (0.9IB corner):

Input Pair:

L = 0.27 μm

W=8.03 μm

VicM=1.1282 V

CM Load:

L = 1.00 μm

W=15.81 μm

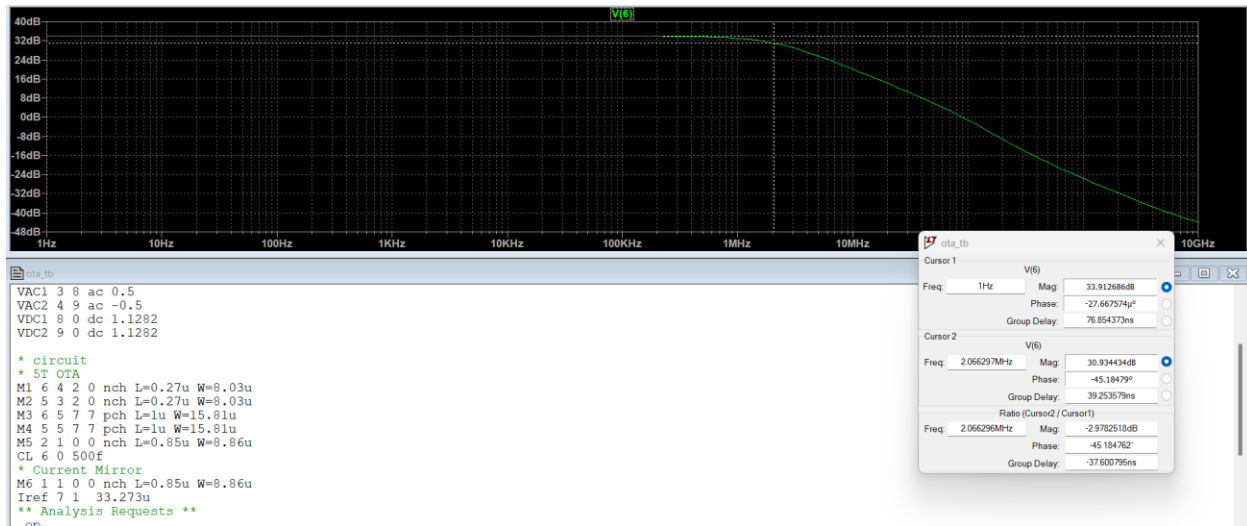
Tail Current Source:

L = 0.85 μm

W=8.86 μm

IB =

3.6973e-05



Achieved the specs at the worst-case corner.

we have more optimized current than local optimization method, we won't run the function across the other corners as it would have exaggerated simulation time.